Khushaan Virk

Vancouver, BC | 236-412-0495 | khushaanvirk@gmail.com | LinkedIn | khushaanvirk.com

Education

University of British Columbia — Vancouver, BC
Bachelor of Applied Science (BASc) in Electrical Engineering — *Expected May 2028*

- **Relevant Coursework:** Circuit Analysis, Digital Logic Design, Signals & Systems, Embedded Systems Design

Experience

Software Engineer — Jan. 2026 – Present
Advizr — *Vancouver, BC*

- Built Python backend services integrating Claude and OpenAI APIs to automate lead generation, content workflows, and operational pipelines for 3 clients across real estate, fintech, and healthcare
- Shipped FastAPI backends and scheduled automation pipelines on Railway with Supabase as the data layer and Vercel-hosted client frontends, eliminating 30+ hours/week of manual workflows across deployed client systems
- Owned client engagements end to end from open-ended business goals to production, including a Canada-wide lead generation engine producing 11,000+ leads with on-demand rescraping

Sports Instructor / Summer Camp Leader — Sep. 2023 – Sep. 2025
City of Surrey — *Surrey, BC*

- Led structured programs for groups of 15–30, coordinating daily operations across staff and volunteers while maintaining safety standards
- Adapted instruction in real time to participants of varying skill levels, communicating clear expectations to children, parents, and co-instructors while resolving conflicts and incidents on the spot

Projects

Field-Following Autonomous Robot | *STM32, C, MOSFETs, I2C, IR* — Apr. 2026

- Built a dual-microcontroller robot with MOSFET H-bridge motor drivers and opto-isolated control lines for electromagnetic noise reduction during inductive sensor readings
- Implemented magnetic field detection using pick-up inductors to autonomously follow a guide wire, with a third sensor for intersection detection and pre-configured path navigation
- Designed an IR remote controller on a separate MCU transmitting 38kHz modulated commands for manual control and path selection
- Integrated VL53L0X Time-of-Flight laser ranging modules over I2C for real-time collision detection and avoidance in autonomous mode

Reflow Oven Controller | *Op-Amp, ADC, SSR, FSM, Python* — Feb. 2026

- Designed a closed-loop temperature control system using a K-type thermocouple with cold-junction compensation (LM335) and OP07 op-amp signal conditioning, achieving ±3°C accuracy over 25–240°C
- Wrote interrupt-driven embedded firmware implementing a 5-state FSM with PWM-based SSR control for precise thermal profile tracking through preheat, soak, reflow, and cooling phases
- Built a Python serial data acquisition dashboard with real-time strip-chart plotting and automated Excel report generation over UART at 115200 baud
- Integrated LCD/7-segment displays, speaker feedback, and safety logic with automatic abort on sensor failure

Battery Temperature Monitor PCB | *KiCad, LM393, NTC Thermistor, Analog Design* — Dec. 2025

- Designed a CR2032-powered analog over-temperature alert: NTC5D-9 thermistor in a ratiometric divider feeding an LM393 comparator with open-collector LED drive (chosen over an op-amp Schmitt trigger for 3V coin-cell compatibility)
- Took the board through the full KiCad flow: schematic capture, 2-layer layout with continuous ground pour, manual analog routing, mixed THT and SMD placement, DRC verification, and Gerber export

AC Phasor Voltmeter | *C, ADC, Signal Processing* — Nov. 2025

- Designed a microcontroller-based AC instrument displaying RMS voltage and phase difference between two 60Hz sinusoidal inputs using high-speed ADC sampling and digital signal processing in C

Technical Skills

Embedded & Control Systems: ARM (STM32), FPGA (DE10-Lite, Nios V soft core), ADC, I2C, SPI, UART, PWM
Programming: C, Embedded C, Python, SystemVerilog, SQL
Instrumentation & Hardware: Oscilloscopes, multimeters, logic analyzers, soldering, signal conditioning, PCB layout
Tools & Software: Git, KiCad, EasyEDA, MATLAB, serial communication (PuTTY), CrossIDE